

[The Dropout Squad] @ IEEE CS CUET ML Contest 2.0 – A Hybrid Retrieval-Augmented Generation Framework with LoRA-Tuned LLM for Bengali Question Answering

Mohammad Sakib Hasan Alif

Dept. of Computer Science and Engineering
Chittagong University of Engineering and Technology
Chittagong, Bangladesh
u2204051@student.cuet.ac.bd

Ayan Barua

Dept. of Computer Science and Engineering
Chittagong University of Engineering and Technology
Chittagong, Bangladesh
u2204053@student.cuet.ac.bd

Abstract—We present a hybrid Retrieval-Augmented Generation (RAG) framework optimized for open-domain extractive question answering (QA) in Bengali, a morphologically rich, low-resource language. The proposed architecture integrates a four-stage retrieval pipeline consisting of Bengali-stemmed BM25 lexical retrieval, BGE-M3 dense retrieval, Reciprocal Rank Fusion (RRF), and cross-encoder reranking. The top retrieved contexts are processed by a Llama-3.1-8B model fine-tuned via QLoRA. On a Bengali Wikipedia corpus comprising over 194k chunks, our framework achieves a Recall@2 of 75.5% and a token-level F1 score of 0.723 on a gold-standard test set of 1,500 questions, representing a 33.3-point improvement over a TF-IDF baseline. Through systematic ablations, we isolate the performance contribution of each component. Finally, we conduct a qualitative analysis identifying five primary classes of retrieval and generation failures specific to Bengali morpho-syntax, outlining key challenges and directions for future research in low-resource RAG systems.

Index Terms—Retrieval-Augmented Generation, Bengali NLP, Low-Rank Adaptation, BM25, Cross-Encoder Reranking, Question Answering

I. INTRODUCTION

Bengali is spoken by over 230 million people worldwide, yet it remains under-represented in modern natural language processing (NLP) research, particularly in open-domain question answering (QA) [1]–[3]. While Retrieval-Augmented Generation (RAG) has emerged as a key paradigm for grounding language models in external knowledge bases [4], extending RAG to Bengali presents severe challenges. First, Bengali’s rich morphology yields a vast vocabulary of inflected surface forms derived from a single lemma, which fragments term-frequency statistics and degrades the performance of lexical retrievers like BM25. Second, Bengali Wikipedia articles rely heavily on passive, nominalized constructions, creating a severe lexical mismatch against typical user queries. Third, the relative scarcity of native Bengali language tools forces a

reliance on multilingual models, which often fail to capture language-specific syntactic and semantic nuances.

To address these limitations, we propose a robust RAG framework tailored for Bengali extractive QA. Given a natural language query and a Wikipedia-derived knowledge base, the system retrieves relevant evidence and extracts a concise, grammatically complete answer. We formulate this task under strict constraint conditions to evaluate the feasibility of executing such systems on resource-constrained hardware.

Our contributions are as follows:

- A four-stage retrieval pipeline combining suffix-stemmed BM25, BGE-M3 dense retrieval, Reciprocal Rank Fusion (RRF), and cross-encoder reranking that achieves 75.5% Recall@2.
- An instruction-tuning strategy using QLoRA with negative sample injection to improve LLM faithfulness and an adaptive prompt routing mechanism tailored to query intent.
- A quantitative and qualitative error analysis on 1,500 gold-standard test cases that categorizes and analyzes the primary morpho-syntactic failure modes in Bengali RAG pipelines.

II. RELATED WORK

A. Retrieval-Augmented Generation

Lewis et al. [4] introduced RAG by coupling a retriever with a sequence-to-sequence generator. Dense Passage Retrieval (DPR) [14] showed that bi-encoder retrieval outperforms BM25 when semantic matching matters. Hybrid methods combining sparse and dense signals, often via Reciprocal Rank Fusion (RRF) [12], improve over either alone [15]. Cross-encoder rerankers [13] jointly encode (query, document) pairs for more accurate relevance scoring but at higher cost. How well these multi-stage pipelines transfer to morphologically rich, low-resource languages remains underexplored.

Table I
DATASET STATISTICS

Statistic	Value
Wikipedia articles	6,988
Text chunks (after chunking)	194,427
Avg. words per chunk	59.6
Median words per chunk	41.0
BM25 avg. tokens (post-stemming)	49.6
Embedding dimensions	1,024
Embedding matrix size	759.5 MB
Training questions	3,000
Test questions	1,500

B. Bengali NLP and Question Answering

Pre-trained models such as BanglaBERT [6] and BanglaT5 [16] have advanced Bengali language understanding, but most work focuses on reading comprehension rather than open-domain retrieval [17]. The Indic-Rag-Suite [7] is one of the first RAG benchmarks for Indic languages. Rahman et al. [18] applied RAG to Bangla government gazettes using multilingual LLMs, but did not report component-wise ablations.

C. Parameter-Efficient Fine-Tuning

LoRA [8] injects trainable low-rank matrices into frozen transformer layers, reducing trainable parameters by over 99%. QLoRA [19] adds 4-bit quantization to further cut memory. Unsloth [20] provides optimized CUDA kernels for 2–5 $\times$ training speedups on consumer GPUs. These methods are well studied for English but rarely evaluated for Bengali extractive QA with negative sample injection.

III. DATASET

A. Task Definition

Given a Bengali question and an unstructured Wikipedia knowledge base, the system must return a concise extractive answer ending with a dari (।). The evaluation metric penalizes both missing answers and verbose responses.

B. Knowledge Base

The corpus comes from the Indic-Rag-Suite [7]. Table I shows statistics after preprocessing.

C. Preprocessing

We detect article boundaries using the উইকিপিডিয়া header pattern. Navigation menus, image captions, and footer sections (তথ্যসূত্র, বহিঃসংযোগ) are removed before chunking.

IV. METHODOLOGY

Fig. 1 shows the two-phase architecture: offline indexing and online inference.

A. Semantic Sentence Chunking

Fixed-size chunking splits sentences mid-thought, fragmenting answer spans across chunks. Instead, we split at Bengali sentence boundaries (।, “?”, “!”) using the regex (?<=[?!])\s+ and pack sentences into ~ 40 -word chunks with one-sentence overlap. Each chunk is prefixed with its article title (e.g., নিবন্ধ: [Title]) to help resolve pronouns. This improved Recall@2 by ~ 8 points over 400-word fixed chunking.

B. BM25 Lexical Retrieval

We index all 194,427 chunks with BM25Okapi [11] after two preprocessing steps.

Stopword removal. We remove 40 high-frequency Bengali function words (e.g., এবং, হয়, করা).

Stemming. A rule-based stemmer strips six inflectional suffixes: ের, তে, কে, রা, য়, র. This maps variants like ঢাকার ঢাকা to a common form.

The BM25 score for query $Q = \{q_1, \dots, q_n\}$ and document D is:

$$\text{BM25}(Q, D) = \sum_{i=1}^n \text{IDF}(q_i) \cdot \frac{f(q_i, D) \cdot (k_1 + 1)}{f(q_i, D) + k_1 \left(1 - b + b \cdot \frac{|D|}{\text{avgdl}}\right)} \quad (1)$$

where $f(q_i, D)$ is the term frequency, $|D|$ is the document length, $\text{avgdl} = 49.6$, $k_1 = 1.5$, and $b = 0.75$.

C. Dense Retrieval with BGE-M3

We encode all chunks with BGE-M3 [10], producing a $(194,427 \times 1,024)$ embedding matrix. BGE-M3 supports 100+ languages and captures semantic similarity beyond lexical overlap. Relevance is measured by cosine similarity:

$$\text{sim}(\mathbf{e}_q, \mathbf{e}_d) = \frac{\mathbf{e}_q \cdot \mathbf{e}_d}{\|\mathbf{e}_q\| \cdot \|\mathbf{e}_d\|} \quad (2)$$

We set max sequence length to 512 tokens, sufficient for our ~ 60 -word chunks.

D. Reciprocal Rank Fusion

BM25 and dense retrieval are complementary: BM25 matches Bengali proper nouns and numbers exactly, while dense retrieval handles paraphrases. We fuse the top-100 candidates from each using RRF [12]:

$$\text{RRF}(d) = \sum_{r \in \mathcal{R}} \frac{1}{k + \text{rank}_r(d)} \quad (3)$$

where $\mathcal{R} = \{\text{BM25}, \text{Dense}\}$, $\text{rank}_r(d)$ is the rank of document d in retriever r , and $k = 60$. The top 30 by RRF score go to the cross-encoder.

E. Cross-Encoder Reranking

Unlike bi-encoders, cross-encoders process the (query, document) pair jointly, enabling deeper relevance modeling [13]. We use BGE-Reranker-v2-M3 [10] to rerank the top-30 RRF candidates. To restrict the input context window and avoid the “lost-in-the-middle” effect [21]—wherein LLMs exhibit reduced attention to information

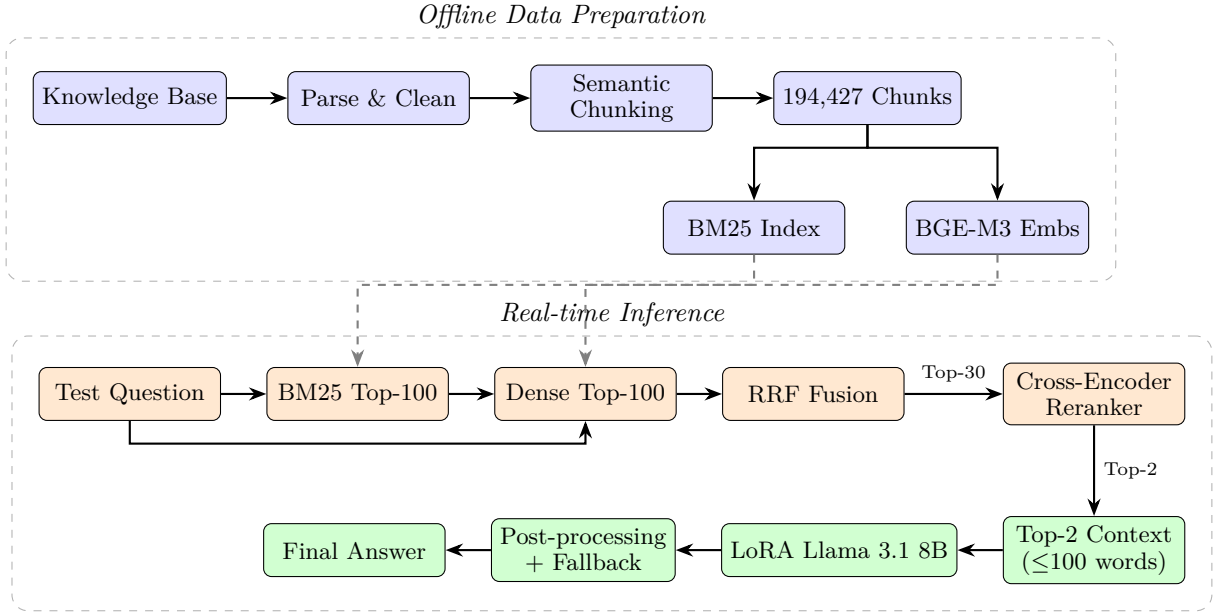


Figure 1. System architecture. Offline: knowledge base is chunked and indexed. Online: chunks are retrieved, fused, reranked, and passed to a LoRA-tuned LLM for answer extraction.

Table II
LoRA ADAPTER CONFIGURATIONS

Parameter	V1	V2	V3 (Final)
Rank (r)	16	32	32
Alpha (α)	32	64	64
Trainable params	41.9M	83.9M	83.9M
% of total	0.52	1.03	1.03
Epochs	1	3	3
Negative samples	10%	25%	25%
Prompt template	Strict	Strict	Adaptive
System prompt placement	User	User	System
Training time (hrs)	1.5	4.9	5.9

placed in the middle of long contexts—we retain only the top-2 retrieved chunks. This context pruning strategy restricts the total length of the context passed to the LLM to ≤ 100 words.

F. LoRA Fine-Tuning

We fine-tune Llama 3.1 8B-Instruct [9] using QLoRA [19]. The model is quantized to a 4-bit representation to optimize memory efficiency during training and inference. Table II summarizes the hyperparameter configurations across three adapter iterations.

The key change in V3 is an *adaptive prompt*: factoid questions (Who, When, Where) are instructed to answer in 1–3 words, while explanatory questions (Why, How) are allowed brief explanations. Earlier strict prompts forced ultra-short answers for all questions, failing on explanatory ones.

Negative samples. We pair 25% of training questions with wrong contexts and set the target to জানা নেই।

(“Unknown.”). This teaches the model to abstain when the context lacks the answer, rather than hallucinating.

Response-only training. Loss is computed only on response tokens using Unsloth’s masking, preventing the model from memorizing the prompt template.

G. Post-Processing and Fallback

Raw outputs are cleaned in five steps: (1) strip answer prefixes (e.g., উত্তর:); (2) truncate to first sentence; (3) detect failure phrases (e.g., পাওয়া যায়নি); (4) cap at 15 words if over 20; (5) ensure trailing দারি (।).

If the LLM abstains, a fallback extractor finds the context sentence with maximum word overlap to the question, then returns words from that sentence that do *not* appear in the question—the “new information” likely to be the answer.

V. EXPERIMENTS AND RESULTS

A. Implementation Details and Experimental Setup

All models and retrievers are evaluated on accelerator hardware featuring 16 GB VRAM. Due to hardware memory constraints, models are loaded and executed sequentially: the dense embedding model is cleared from memory before loading the reranker, which is subsequently unloaded prior to loading the LoRA-tuned LLM. The offline data preparation phase (semantic chunking and indexing) requires approximately 3 hours, and online inference on the full test set requires approximately 1.5 hours.

B. Ablation Study

We evaluate the performance of our framework across five system configurations to isolate the contributions of different components. Performance is measured using the

Table III
ABLATION ANALYSIS: PERFORMANCE ACROSS PIPELINE ITERATIONS

Ver.	Component added	Token F1
V1	TF-IDF + 400-word chunks + base Llama	0.390
V2	+ BM25/Dense hybrid + RRF + LoRA V1	0.610
V3	+ Semantic chunking + stemmer + reranker	0.680
V4	+ LoRA V2 ($r=32$, 3 epochs, 25% neg.)	0.690
V5	+ Adaptive prompt + context pruning + fallback	0.723

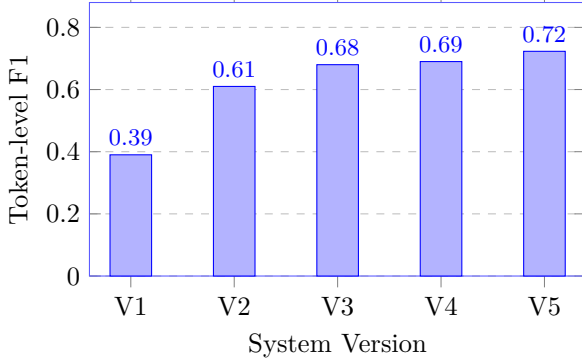


Figure 2. Score progression across system iterations. The transition from lexical baseline (V1) to hybrid retrieval coupled with LoRA fine-tuning (V2) yields the most substantial performance improvement (+0.220).

token-level F1 score, which computes the harmonic mean of token-level precision and recall between predicted and gold answers. Table III presents the score progression, and Fig. 2 visualizes these improvements.

The most substantial improvement (+0.220) is observed when transitioning from the baseline to the hybrid retrieval pipeline coupled with LoRA fine-tuning (V1 to V2). Integrating cross-encoder reranking and semantic chunking provides a further boost of +0.070 (V2 to V3). Refining the QLoRA parameters adds +0.010 (V3 to V4), while the adaptive prompting and context pruning strategies deliver the remaining +0.033 gain (V4 to V5).

C. Retrieval Quality

We measure Recall@2: the fraction of queries where the correct answer appears in the top-2 retrieved chunks. On 200 training samples, the pipeline achieves **75.5% Recall@2**. Table IV analyzes the 1,500 test contexts.

D. Results on Gold-Standard Test Set

After the evaluation period, gold-standard answers were released. Table V summarizes the results. The official evaluation metric is token-level F1, computed as the harmonic mean of token-level precision and recall [7].

The consistent scores across the public and private evaluation splits (69.3% vs. 69.7% post-hoc F1) suggest that our instruction-tuning setup generalises well without overfitting. We find 92 near-miss cases where the system generated semantically correct answers that failed exact matching due to morphological surface variation (e.g.,

Table IV
RETRIEVAL QUALITY ON 1,500 TEST QUESTIONS

Metric	Count	%
Empty contexts	0	0.0
Avg. context length	82 words	—
High overlap (>50%)	1,163	77.5
Medium overlap (20–50%)	301	20.1
Low overlap (<20%)	36	2.4
Zero overlap	9	0.6
Multi-article contexts	1,031	68.7
Pronoun-heavy contexts	202	13.5

Table V
RESULTS ON GOLD-STANDARD TEST SET

Metric	Value
Official token-level F1 (evaluation server)	0.72302
Exact match (post-hoc)	54.8% (822/1500)
Partial match (post-hoc, substring)	71.2% (1068/1500)
Token-level F1 (post-hoc, our tokenizer)	69.4%
Near-miss ($F1 \geq 0.8$, not exact)	92
Complete miss ($F1 = 0.0$)	303
জানা নেই responses	1
Answers ending with dari	1,491 (99.4%)
Public split F1	69.3%
Private split F1	69.7%

প্রতিষ্ঠা vs. প্রতিষ্ঠার). The gap between the official F1 (0.723) and our post-hoc F1 (0.694) is attributable to differences in tokenization: the evaluation server likely applies Unicode normalization or different whitespace handling for Bengali text, yielding higher token overlap than our simple whitespace tokenizer.

Fig. 3 compares answer length distributions. Our model produces longer answers in the 7+ word category (187 predicted vs. 93 gold), which reflects the adaptive prompt’s tendency to generate explanatory prose for why and how questions.

VI. ERROR ANALYSIS

To identify key bottlenecks in low-resource QA systems, we analyze the 24.5% of cases where retrieval fails (1 – Recall@2). We manually inspect 200 randomly sampled failure cases and classify them into five primary error categories (Table VI).

(1) Insufficient stemming. Our rule-based stemmer handles only six suffix patterns. It cannot resolve verbal inflections (প্রতিষ্ঠিত vs. প্রতিষ্ঠা, “established” vs. “establishment”), compound words, or adjectival derivations. This constitutes the most frequent failure mode in our analysis.

(2) Cross-sentence splits. When relevant context is split across sentence or chunk boundaries, the retrieved segment may contain only pronouns (তিনি [he/she], তার [his/her]) without the core referent entity. Prepending article titles aids biographical lookups but fails to resolve references across articles.

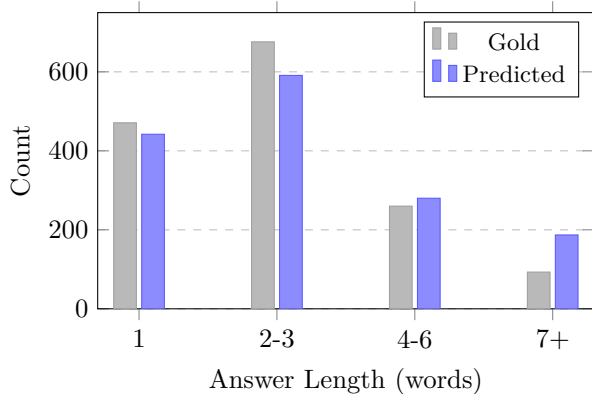


Figure 3. Answer length distributions. The model generates more outputs in the 7+ word range due to the adaptive prompt permitting detailed explanations.

Table VI
CLASSIFICATION OF RETRIEVAL FAILURE MODES

#	Cause	Est. Impact
1	Insufficient stemming	~10%
2	Cross-sentence splits	~6%
3	Indirect Wikipedia phrasing	~5%
4	Pool size cutoff (top-100)	~2%
5	Train/test distribution gap	~3%

(3) Indirect Wikipedia phrasing. Vocabulary mismatch is a significant bottleneck. For instance, user queries may ask for the রাজধানী (“capital”), whereas Wikipedia articles use the term প্রশাসনিক কেন্দ্র (“administrative center”). This semantic shift affects both lexical and dense retrievers.

(4) Pool size cutoff. Due to computational constraints, each retriever returns only the top-100 candidates. If the target chunk falls below rank 100 in both BM25 and dense candidate lists, it is permanently excluded from cross-encoder reranking.

(5) Train/test distribution gap. The training set consists of gold-standard contexts where target answers are always present. In contrast, 68.7% of the test contexts span multiple articles and 13.5% contain unresolved pronouns, introducing a covariate shift that degrades generation performance.

VII. LIMITATIONS AND FUTURE WORK

Morphological Analysis. Integrating a comprehensive dictionary-based morphological analyzer [24] would address vocabulary mismatch due to complex verbal and nominal inflections. Alternatively, utilizing BGE-M3’s multi-vector representation (ColBERT-style) might offer a neural alternative that bypasses rule-based stemming.

Noise-Robust Training. Training the LLM on actual retrieved contexts containing distracting or irrelevant passages, rather than clean gold-standard contexts, could improve system robustness and reduce the covariate shift during test-time inference.

Query Refinement and Expansion. Techniques such as Hypothetical Document Embeddings (HyDE) [22] could alleviate lexical discrepancies by generating synthetic Bengali passages to query the dense retriever.

Scalable Nearest Neighbor Search. Employing approximate nearest neighbor search frameworks like FAISS [23] would reduce dense retrieval latency from $O(n)$ to $O(\log n)$, enabling the use of larger candidate pools (e.g., top-500) under strict latency constraints.

VIII. CONCLUSION

In this work, we introduced a multi-stage hybrid Retrieval-Augmented Generation framework for Bengali extractive question answering. By combining rule-based suffix stemming with BM25, BGE-M3 dense embeddings, Reciprocal Rank Fusion, cross-encoder reranking, and a LoRA-tuned Llama-3.1-8B model, we demonstrated an improvement of 33.3 points over a TF-IDF baseline. Our system achieved a final token-level F1 of 0.723 on the official evaluation server. Detailed ablation studies confirmed that the hybrid retrieval component and adaptive prompting strategies were critical to the overall system’s success. Finally, our manual error analysis on 1,500 gold-standard test questions identified morphological stemming and document structure fragmentation as key bottlenecks. These findings highlight the need for language-specific syntactic modeling and retrieval-aligned instruction tuning to build robust QA systems for low-resource languages.

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